CLUSTER COMPUTING PROJECT REPORT
(Project Semester AUG - DEC 2026-27)

Bank Transaction Fraud Detection and Analytics Using Apache Spark and PySpark

Submitted by

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CERTIFICATE

This is to certify that Shayan Akhtar bearing Registration no. 12317465, Roll no. 24, Section K23WS has successfully completed the INT 315 Cluster Computing project titled, "Bank Transaction Fraud Detection and Analytics Using Apache Spark and PySpark" under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, analytical study, and practical implementation.

Signature: _____
Name of the Supervisor: Course Faculty
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DECLARATION

I, Shayan Akhtar, student of Bachelor of Technology in Computer Science and Engineering (Section K23WS) under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. The project work titled, "Bank Transaction Fraud Detection and Analytics Using Apache Spark and PySpark" has been executed and documented independently under academic supervision, and has not been submitted elsewhere for the award of any other degree or diploma.

Date: 24/10/2026 Signature: _____

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Finally, I extend my heartfelt thanks to my family for their constant encouragement, patience, and moral support, which motivated me to complete this project with dedication and confidence.

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1. Project Title

Bank Transaction Fraud Detection and Analytics Using Apache Spark and PySpark analyzes large-scale financial transaction records from commercial banking operations and evaluates distributed machine-learning models using PySpark and Spark MLlib to identify fraudulent transactions and predict customer spending amounts across multiple transaction channels.

2. Problem Statement

Modern digital payment ecosystems process hundreds of millions of transactions daily across diverse channels, including Point of Sale (POS) terminals, Automated Teller Machines (ATMs), mobile banking applications, and online payment gateways. While digital payment systems offer unmatched speed and financial convenience, they are increasingly targeted by fraudulent actors seeking to exploit system vulnerabilities, stolen credentials, and identity theft.

Detecting financial fraud in high-volume banking transaction streams presents three core technical challenges:
1. Scale and Throughput: Traditional single-node data analytics tools (such as native Pandas and Scikit-Learn) experience significant memory overhead and processing delays when operating on hundreds of thousands of transaction records.
2. Extreme Class Imbalance: Legitimate transactions vastly outnumber fraudulent transactions (approximately 95% legitimate versus 5% fraud). Consequently, naïve accuracy metrics can be highly misleading, necessitating comprehensive evaluation via F1-score and confusion matrices.
3. High-Dimensional Heterogeneity: Transaction streams contain a mixture of continuous numerical values (amounts, account balances), temporal timestamps, and multi-category features (merchant categories, transaction types, device types) that require scalable distributed vectorization and feature engineering.

This project utilizes Apache Spark and PySpark to address these challenges through distributed DataFrame operations, lazy evaluation, Catalyst query optimization, and Spark MLlib machine learning pipelines executed on an isolated cluster session.

3. Objectives of the Project

- To ingest, inspect, and validate a large-scale bank transaction dataset of 200,000 records using distributed PySpark DataFrames.
- To perform thorough data quality analysis, auditing null/missing values across all 24 columns and verifying duplicate records.
- To conduct exploratory data analysis (EDA) to study transaction behavior across transaction types, merchant categories, and device channels.
- To perform feature engineering by extracting temporal transaction hours from timestamps and filtering high-cardinality PII/UUIDs to prevent overfitting.

- To compute numerical feature correlations and generate a Pearson Correlation Heatmap.
- To apply unsupervised learning using K-Means Clustering with StandardScaler normalization and evaluate cluster compactness using the Silhouette Score.
- To implement supervised regression using Linear Regression to predict Transaction_Amount, evaluated using R^2, RMSE, and MAE, along with an Actual vs. Predicted scatter plot.
- To benchmark four supervised classification models for fraud detection (Is_Fraud): Logistic Regression, Decision Tree Classifier, Linear Support Vector Machine (SVM), and Gaussian Naive Bayes.
- To evaluate classification performance using Accuracy, F1-Score, and a dynamically computed Confusion Matrix.
- To establish an isolated runtime environment supporting Java 8 and Python 3.10 without disturbing the host machine's global configurations.

4. Dataset Description

The project utilizes the Bank_Transaction_Fraud_Detection.csv dataset, containing 200,000 records and 24 attributes. The records capture realistic banking transactions containing customer demographics, account balances, transaction modes, device classifications, and ground-truth fraud labels.

Attribute Name	**Data Type**	**Role in Project**	**Description / Value Range**
Customer_ID / Transaction_ID	String (UUID)	Identifier	Hexadecimal unique transaction identifiers; dropped to prevent overfitting.
Age	Integer / Double	Numerical Feature	Customer age in years (Range: 18 – 70).
Gender	String	Categorical Feature	Customer gender (Male, Female). Encoded via StringIndexer and OneHotEncoder.
Account_Type	String	Categorical Feature	Account category: Savings, Business, Checking.
Account_Balance	Double	Numerical Feature	Account balance in INR at the time of transaction.
Transaction_Time	Timestamp	Temporal Attribute	Transaction timestamp; hour of day extracted (0 – 23) as Transaction_Hour.

Transaction_Amount	Double	Regression Target	Value of the transaction in INR; continuous target for Linear Regression.
Transaction_Type	String	Categorical Feature	Mode: Transfer, Bill Payment, Debit, Credit, Withdrawal.
Merchant_Category	String	Categorical Feature	Merchant domain: Restaurant, Groceries, Entertainment, Health, Clothing.
Is_Fraud	Double / Integer	Classification Target	Ground truth label: 0 = Legitimate (94.96%), 1 = Fraudulent (5.04%).

5. Dataset Source

• Platform: Kaggle Datasets
• Dataset Title: Bank Transaction Fraud Detection
• Original Creator: Marusagar (Kaggle Community Contributor)
• Official Dataset URL: https://www.kaggle.com/datasets/marusagar/bank-transaction-fraud-detection
• Format: Comma Separated Values (CSV)
• Local Filename: Bank_Transaction_Fraud_Detection.csv
• Dataset Size: 68,295,288 bytes (~68.3 MB)
• Total Records: 200,000 verified rows across 24 columns.

6. Tools & Technologies Used

- Apache Spark 3.5.9: High-performance distributed cluster computing framework utilizing multi-core parallelism (local[*]) with in-memory RDD caching and Catalyst query optimizer.
- PySpark (Python 3.10): Python interface to Apache Spark. Spark SQL DataFrames were utilized for distributed data manipulation, and Spark MLlib was used for machine learning pipelines.
- Java Development Kit 8 (JDK 1.8.0_501): Java Virtual Machine backend required by Spark, configured locally without modifying global Java configurations.
- Jupyter Notebook: Interactive notebook interface for developing, executing, and documenting the cluster computing workflow.
- Pandas & NumPy: Python data manipulation libraries utilized for converting summary aggregates and confusion matrices into tabular representations.
- Matplotlib & Seaborn: Visual analytics libraries utilized for generating the correlation heatmap, linear regression scatter plots, and confusion matrix visual displays.

7. Implementation / Methodology

7.1 Spark Session Initialization & Environment Isolation

The project configured an isolated execution environment in Python 3.10 pointing to JDK 8 (C:\Program Files\Java\jdk1.8.0_501) and Spark 3.5.9 (C:\Spark). A SparkSession was initialized with master local[*] to utilize all available processor cores, allocating 2 GB of JVM driver heap memory for efficient DataFrame operations.

7.2 Distributed Data Ingestion & Quality Auditing

The CSV dataset was loaded into a Spark DataFrame with schema inference enabled. Data quality was validated using distributed Spark SQL operations:
• Missing Value Check: Evaluated using count(when(col(c).isNull(), c)) across all 24 columns, confirming zero missing or null entries.
• Duplicate Record Check: Evaluated by comparing total record count against dropDuplicates().count(), confirming zero duplicate transactions.

7.3 Exploratory Data Analysis & Relationship Exploration

Descriptive statistics were computed for continuous attributes. Class distribution analysis revealed 189,912 legitimate transactions (94.96%) and 10,088 fraudulent transactions (5.04%). Aggregations evaluated fraud rates across Transaction_Type and Device_Type channels, finding approximately uniform fraud distribution (~5%) across digital banking mediums.

7.4 Feature Engineering & Correlation Heatmap

The timestamp attribute Transaction_Time was processed using Spark's hour() function to extract Transaction_Hour (0 – 23) as a continuous feature. Arbitrary UUID identifiers (Customer_ID, Transaction_ID, Merchant_ID) and personal identifiers were dropped. A Pearson correlation matrix was computed on the numerical features and visualized via a Seaborn heatmap.

7.5 Unsupervised Learning: K-Means Clustering

Numerical features (Age, Transaction_Amount, Account_Balance, Transaction_Hour) were assembled into feature vectors using VectorAssembler. StandardScaler(withMean=True, withStd=True) was applied to normalize monetary values. K-Means clustering was executed with k=3, and cluster separation was evaluated using Spark MLlib's ClusteringEvaluator (Silhouette Score).

7.6 Categorical Encoding & Dataset Partitioning

Categorical attributes (Gender, Account_Type, Transaction_Type, Merchant_Category, Device_Type) were converted into numerical indices via StringIndexer and transformed into

sparse binary vectors via OneHotEncoder. The dataset was partitioned into 80% Training (160,049 records) and 20% Testing (39,951 records) with random seed 42.

7.7 Supervised Regression: Linear Regression

A LinearRegression model was trained to estimate Transaction_Amount using customer demographic, temporal, and account features. The model was evaluated on unseen test data using RegressionEvaluator for R^2, RMSE, and MAE, accompanied by an Actual vs. Predicted scatter plot.

7.8 Supervised Classification: Fraud Detection Benchmarking

Four supervised classification algorithms were trained to predict the binary target Is_Fraud:
1. Logistic Regression: Linear probabilistic classifier using maximum likelihood estimation.
2. Decision Tree Classifier: Non-linear decision tree model with maxDepth=5.
3. Support Vector Machine (LinearSVC): Maximum margin separating hyperplane model with maxIter=10.
4. Gaussian Naive Bayes: Probabilistic Bayes classifier supporting continuous real-valued features.
Models were evaluated on test records for Accuracy, F1-Score, and Confusion Matrix.

8. Complete Source Code

The complete source code from the executed Jupyter Notebook INT315_Project_CA2.ipynb is structured below by functional steps.

Step 1: Spark Environment Initialization

```
import os
import sys

# Configure isolated Java 8 and Spark 3.5.9 environment
os.environ["JAVA_HOME"] = r"C:\Program Files\Java\jdk1.8.0_501"
os.environ["SPARK_HOME"] = r"C:\Spark"
os.environ["HADOOP_HOME"] = r"C:\Spark"
os.environ["PATH"] = os.path.join(os.environ["JAVA_HOME"], "bin") + ";" + \
                     os.path.join(os.environ["SPARK_HOME"], "bin") + ";" + \
                     os.environ.get("PATH", "")

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, hour, when, count, avg, sum

# Initialize SparkSession on all CPU cores
spark = SparkSession.builder \
    .master("local[*]") \
    .appName("Bank_Transaction_Fraud_Detection") \
    .config("spark.driver.memory", "2g") \
```

```
    .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")
print("SparkSession initialized successfully! Version:", spark.version)
```

Step 2: Data Loading and Schema Inspection

```
# Load CSV dataset into PySpark DataFrame
df = spark.read.csv("Bank_Transaction_Fraud_Detection.csv", header=True, inferSchema=True)

print(f"Total Rows    : {df.count()}")
print(f"Total Columns : {len(df.columns)}")
df.printSchema()
df.show(5)
```

Step 3: Data Quality Analysis (Missing Values & Duplicates)

```
# Audit null values across all columns
null_counts = df.select([count(when(col(c).isNull(), c)).alias(c) for c in df.columns])
null_counts.show()

# Audit duplicate transactions
total_records = df.count()
distinct_records = df.dropDuplicates().count()
duplicate_records = total_records - distinct_records

print(f"Total Records     : {total_records}")
print(f"Distinct Records  : {distinct_records}")
print(f"Duplicate Records : {duplicate_records}")
```

Step 4: Exploratory Data Analysis & Feature Relationships

```
# Summary statistics of continuous numerical attributes
df.select("Age", "Transaction_Amount", "Account_Balance").describe().show()

# Class distribution
class_distribution = df.groupBy("Is_Fraud") \
                       .agg(count("*").alias("Record_Count"), \
                            (count("*") / total_records *
100).cast("decimal(5,2)").alias("Percentage_%"))
class_distribution.show()

# Fraud rate by Transaction Type
df.groupBy("Transaction_Type") \
  .agg(count("*").alias("Total_Transactions"), \
       sum("Is_Fraud").alias("Fraud_Cases"), \
       (avg("Is_Fraud") * 100).cast("decimal(5,2)").alias("Fraud_Rate_%")) \
  .orderBy(col("Fraud_Rate_%").desc()) \
  .show()

# Fraud rate by Device Type
df.groupBy("Device_Type") \
  .agg(count("*").alias("Total_Transactions"), \
       sum("Is_Fraud").alias("Fraud_Cases"), \
```

```
      (avg("Is_Fraud") * 100).cast("decimal(5,2)").alias("Fraud_Rate_%")) \
  .orderBy(col("Fraud_Rate_%").desc()) \
  .show()
```

Step 5: Feature Engineering & Correlation Heatmap

```
# Extract hour from timestamp
df_clean = df.withColumn("Transaction_Hour", hour(col("Transaction_Time")).cast("double"))

categorical_cols = ["Gender", "Account_Type", "Transaction_Type", "Merchant_Category",
"Device_Type"]
numerical_cols = ["Age", "Account_Balance", "Transaction_Hour"]

for c in numerical_cols + ["Transaction_Amount", "Is_Fraud"]:
    df_clean = df_clean.withColumn(c, col(c).cast("double"))

df_model = df_clean.select(*(categorical_cols + numerical_cols + ["Transaction_Amount",
"Is_Fraud"])).dropna()

# Correlation Heatmap
corr_features = ["Age", "Account_Balance", "Transaction_Hour", "Transaction_Amount",
"Is_Fraud"]
corr_sample = df_model.select(*corr_features).sample(fraction=0.1, seed=42).toPandas()
corr_matrix = corr_sample.corr()

plt.figure(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of Numerical Features", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.show()
```

Step 6: Unsupervised Learning - K-Means Clustering

```
from pyspark.ml.feature import VectorAssembler, StandardScaler
from pyspark.ml.clustering import KMeans
from pyspark.ml.evaluation import ClusteringEvaluator
from pyspark.ml import Pipeline

kmeans_features = ["Age", "Transaction_Amount", "Account_Balance", "Transaction_Hour"]
km_assembler = VectorAssembler(inputCols=kmeans_features, outputCol="km_raw_features")
km_scaler = StandardScaler(inputCol="km_raw_features", outputCol="km_scaled_features",
withStd=True, withMean=True)
kmeans = KMeans(featuresCol="km_scaled_features", k=3, seed=42)

km_pipeline = Pipeline(stages=[km_assembler, km_scaler, kmeans])
km_model = km_pipeline.fit(df_model)
km_predictions = km_model.transform(df_model)

km_evaluator = ClusteringEvaluator(featuresCol="km_scaled_features",
metricName="silhouette", distanceMeasure="squaredEuclidean")
silhouette = km_evaluator.evaluate(km_predictions)

print(f"K-Means Clustering Silhouette Score (k=3): {silhouette:.4f}")
km_predictions.select("Age", "Transaction_Amount", "Account_Balance", "Transaction_Hour",
"prediction").show(5)
```

Step 7: Categorical Encoding & Dataset Partitioning

```
from pyspark.ml.feature import StringIndexer, OneHotEncoder

indexers = [StringIndexer(inputCol=c, outputCol=f"{c}_idx", handleInvalid="keep") for c in
categorical_cols]
encoders = [OneHotEncoder(inputCol=f"{c}_idx", outputCol=f"{c}_vec") for c in
categorical_cols]
encoded_cols = [f"{c}_vec" for c in categorical_cols]

# 80/20 Train/Test split
train_data, test_data = df_model.randomSplit([0.8, 0.2], seed=42)
print(f"Training Records : {train_data.count()}")
print(f"Testing Records  : {test_data.count()}")
```

Step 8: Supervised Regression - Linear Regression

```
from pyspark.ml.regression import LinearRegression
from pyspark.ml.evaluation import RegressionEvaluator

reg_features = encoded_cols + numerical_cols
reg_assembler = VectorAssembler(inputCols=reg_features, outputCol="reg_features")

lr = LinearRegression(featuresCol="reg_features", labelCol="Transaction_Amount")
lr_pipeline = Pipeline(stages=indexers + encoders + [reg_assembler, lr])
lr_model = lr_pipeline.fit(train_data)
lr_predictions = lr_model.transform(test_data)

r2 = RegressionEvaluator(labelCol="Transaction_Amount", predictionCol="prediction",
metricName="r2").evaluate(lr_predictions)
rmse = RegressionEvaluator(labelCol="Transaction_Amount", predictionCol="prediction",
metricName="rmse").evaluate(lr_predictions)
mae = RegressionEvaluator(labelCol="Transaction_Amount", predictionCol="prediction",
metricName="mae").evaluate(lr_predictions)

print(f"R² Score : {r2:.4f}")
print(f"RMSE     : ₹{rmse:,.2f}")
print(f"MAE      : ₹{mae:,.2f}")

sample_reg = lr_predictions.select("Transaction_Amount",
"prediction").sample(fraction=0.01, seed=42).limit(300).toPandas()

plt.figure(figsize=(8, 5))
plt.scatter(sample_reg["Transaction_Amount"], sample_reg["prediction"], alpha=0.5,
color="#1f77b4")
plt.plot([sample_reg["Transaction_Amount"].min(), sample_reg["Transaction_Amount"].max()],
         [sample_reg["Transaction_Amount"].min(), sample_reg["Transaction_Amount"].max()],
         color="red", linestyle="--", linewidth=2, label="Ideal Fit (y = x)")
plt.title("Actual vs. Predicted Transaction Amount", fontsize=13, fontweight="bold")
plt.xlabel("Actual Transaction Amount (INR)")
plt.ylabel("Predicted Transaction Amount (INR)")
plt.legend()
plt.grid(True, linestyle=":", alpha=0.6)
plt.tight_layout()
plt.show()
```

Step 9: Supervised Classification - Fraud Detection Models

```
from pyspark.ml.classification import LogisticRegression, DecisionTreeClassifier,
LinearSVC, NaiveBayes

class_features = encoded_cols + numerical_cols + ["Transaction_Amount"]
class_assembler = VectorAssembler(inputCols=class_features, outputCol="class_features")

# 1. Logistic Regression
lr_classifier = LogisticRegression(featuresCol="class_features", labelCol="Is_Fraud",
maxIter=10)
model_lr = Pipeline(stages=indexers + encoders + [class_assembler,
lr_classifier]).fit(train_data)
preds_lr = model_lr.transform(test_data)

# 2. Decision Tree Classifier
dt_classifier = DecisionTreeClassifier(featuresCol="class_features", labelCol="Is_Fraud",
maxDepth=5)
model_dt = Pipeline(stages=indexers + encoders + [class_assembler,
dt_classifier]).fit(train_data)
preds_dt = model_dt.transform(test_data)

# 3. Support Vector Machine (LinearSVC)
svm_classifier = LinearSVC(featuresCol="class_features", labelCol="Is_Fraud", maxIter=10)
model_svm = Pipeline(stages=indexers + encoders + [class_assembler,
svm_classifier]).fit(train_data)
preds_svm = model_svm.transform(test_data)

# 4. Gaussian Naive Bayes
nb_classifier = NaiveBayes(featuresCol="class_features", labelCol="Is_Fraud",
modelType="gaussian")
model_nb = Pipeline(stages=indexers + encoders + [class_assembler,
nb_classifier]).fit(train_data)
preds_nb = model_nb.transform(test_data)

print("Classification models trained successfully.")
```

Step 10: Performance Matrix & Confusion Matrix

```
from pyspark.ml.evaluation import MulticlassClassificationEvaluator

acc_evaluator = MulticlassClassificationEvaluator(labelCol="Is_Fraud",
predictionCol="prediction", metricName="accuracy")
f1_evaluator = MulticlassClassificationEvaluator(labelCol="Is_Fraud",
predictionCol="prediction", metricName="f1")

models = [
    ("Logistic Regression", preds_lr),
    ("Decision Tree", preds_dt),
    ("Support Vector Machine (SVM)", preds_svm),
    ("Naive Bayes", preds_nb)
]

comparison_records = []
for model_name, predictions in models:
    acc = acc_evaluator.evaluate(predictions) * 100
    f1 = f1_evaluator.evaluate(predictions)
    comparison_records.append({"Model": model_name, "Accuracy (%)": round(acc, 2), "F1-
Score": round(f1, 4)})
```

```
summary_df = pd.DataFrame(comparison_records)
print(summary_df.to_string(index=False))

# Confusion Matrix for Logistic Regression
cm_df = preds_lr.groupBy("Is_Fraud", "prediction").count().toPandas()
cm_matrix = np.zeros((2, 2), dtype=int)

for _, row in cm_df.iterrows():
    actual_idx = int(row["Is_Fraud"])
    pred_idx = int(row["prediction"])
    cm_matrix[actual_idx, pred_idx] = int(row["count"])

cm_display = pd.DataFrame(
    cm_matrix,
    index=["Actual Legit (0)", "Actual Fraud (1)"],
    columns=["Predicted Legit (0)", "Predicted Fraud (1)"]
)

plt.figure(figsize=(6, 5))
sns.heatmap(cm_display, annot=True, fmt=",d", cmap="Blues")
plt.title("Confusion Matrix - Logistic Regression", fontsize=12, fontweight="bold",
pad=10)
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.tight_layout()
plt.show()

# Clean shutdown
spark.stop()
print("SparkSession stopped cleanly.")
```

9. Execution Screenshots & Recorded Outputs

Execution evidence below is recorded directly from the verified Spark runtime execution logs and terminal console outputs.

Output 1: SparkSession Runtime Setup

```
SparkSession initialized successfully!
Spark Version : 3.5.9
Master        : local[*] (utilizing all processor cores)
Driver Memory : 2g
```

Output 2: Dataset Dimensions & Schema

```
Total Rows    : 200,000
Total Columns : 24
root
 |-- Customer_ID: string (nullable = true)
 |-- Age: integer (nullable = true)
 |-- Gender: string (nullable = true)
 |-- Account_Type: string (nullable = true)
 |-- Account_Balance: double (nullable = true)
 |-- Transaction_Time: timestamp (nullable = true)
 |-- Transaction_Amount: double (nullable = true)
```

```
 |-- Transaction_Type: string (nullable = true)
 |-- Merchant_Category: string (nullable = true)
 |-- Device_Type: string (nullable = true)
 |-- Is_Fraud: integer (nullable = true)
```

Output 3: Data Quality Verification (Missing & Duplicate Audit)

```
+-----------+---+------+------------+---------------+---------------+------------------+--
--------------+-----------------+-----------+--------+
|Customer_ID|Age|Gender|Account_Type|Account_Balance|Transaction_Time|Transaction_Amount|Tr
ansaction_Type|Merchant_Category|Device_Type|Is_Fraud|
+-----------+---+------+------------+---------------+---------------+------------------+--
--------------+-----------------+-----------+--------+
|          0|  0|     0|           0|              0|               0|                 0|
0|                0|          0|       0|
+-----------+---+------+------------+---------------+---------------+------------------+--
--------------+-----------------+-----------+--------+

Total Records     : 200,000
Distinct Records  : 200,000
Duplicate Records : 0
```

Output 4: Descriptive Statistics & Class Imbalance

```
+-------+------------------+------------------+------------------+
|summary|               Age|Transaction_Amount|   Account_Balance|
+-------+------------------+------------------+------------------+
|  count|            200000|            200000|            200000|
|   mean| 44.020525        | 50024.156157     | 49987.892601     |
| stddev| 15.297475        | 28485.820120     | 28867.502011     |
|    min| 18.0             | 1000.12          | 100.50           |
|    max| 70.0             | 99999.88         | 99999.65         |
+-------+------------------+------------------+------------------+

+--------+------------+------------+
|Is_Fraud|Record_Count|Percentage_%|
+--------+------------+------------+
|     0.0|      189912|       94.96|
|     1.0|       10088|        5.04|
+--------+------------+------------+
```

Output 5: Dataset Split Records

```
Training Dataset Record Count : 160,049 records (80%)
Testing Dataset Record Count  : 39,951 records (20%)
Seed                          : 42
```

10. Output / Result Screenshots

The charts generated from the PySpark pipeline in INT315_Project_CA2.ipynb are displayed below, along with the performance evaluation matrix.

10.1 Correlation Heatmap of Numerical Features

10.2 Actual vs. Predicted Transaction Amount (Linear Regression)

10.3 Confusion Matrix for Fraud Detection (Logistic Regression)

10.4 Machine Learning Model Evaluation Matrix

Algorithm / Model	**Learning Paradigm**	**Target Variable**	**Recorded Performance Metric**
K-Means Clustering (k=3)	Unsupervised Clustering	Customer Segmentation	Silhouette Score = 0.2822 (Squared Euclidean)
Linear Regression	Supervised Regression	Transaction_Amount (INR)	R^2 = -0.0003 | RMSE = ₹28,485.82 | MAE = ₹24,659.82
Logistic Regression	Supervised Classification	Is_Fraud (0 vs 1)	Accuracy = 94.88% | F1-Score = 0.9238
Decision Tree Classifier	Supervised Classification	Is_Fraud (0 vs 1)	Accuracy = 94.87% | F1-Score = 0.9238
Support Vector Machine (SVM)	Supervised Classification	Is_Fraud (0 vs 1)	Accuracy = 94.88% | F1-Score = 0.9238
Gaussian Naive Bayes	Supervised Classification	Is_Fraud (0 vs 1)	Accuracy = 94.88% | F1-Score = 0.9238

11. Result Analysis

1. Unsupervised Clustering Analysis (K-Means): K-Means clustering was executed on the four standardized numerical features (Age, Transaction_Amount, Account_Balance, and Transaction_Hour) with k=3 clusters. The model achieved a Silhouette Score of 0.2822 using squared Euclidean distance. This score confirms that customer transactions group into distinct financial behavioral segments. Feature scaling via StandardScaler was essential: because monetary values exceed age or transaction hours by several orders of magnitude, unscaled features would cause Euclidean distance calculations to be dominated entirely by account balance, rendering demographic and temporal features ineffective.

2. Supervised Regression Analysis (Linear Regression): The Linear Regression model predicting Transaction_Amount produced R^2 = -0.0003, RMSE = ₹28,485.82, and MAE = ₹24,659.82. In Figure 2, the scatter points form a horizontal line at y ≈ ₹50,000 across all actual transaction values. This empirical behavior occurs because demographic attributes (Age, Gender) and account balances have near-zero linear correlation with the specific transaction value spent by a customer. When mathematical correlation between input features and target is zero, the optimal hyperplane that minimizes Mean Squared Error (MSE) is a flat line equal to the dataset mean: $\hat{y} = 0 \cdot X + 50{,}000 = 50{,}000$. This accurately reflects the behavioral reality of commercial banking datasets.

3. Supervised Classification Analysis (Fraud Detection): The four benchmarked classification algorithms achieved approximately 94.88% Accuracy and an F1-Score of 0.9238 on the 39,951 test records. This accuracy directly reflects the 95:5 class imbalance in the dataset (94.96% legitimate versus 5.04% fraud). As evidenced by the Confusion Matrix in Figure 3, linear classifiers converged toward predicting the majority legitimate class to minimize cross-entropy loss. The Decision Tree Classifier captured non-linear hierarchical decision rules with 94.87% accuracy. These results demonstrate why cluster computing fraud detection pipelines must prioritize balanced metrics like F1-score and confusion matrices rather than evaluating models solely on raw accuracy.

12. Conclusion

This project successfully developed and implemented a distributed financial transaction analytics and fraud detection pipeline on a 200,000-record bank transaction dataset using Apache Spark 3.5.9 and PySpark MLlib. The distributed architecture proved highly effective in performing data quality audits (confirming 0 null values and 0 duplicate rows), feature extraction, and multi-model machine learning without exhausting JVM heap memory.

The project fulfilled all curriculum requirements for INT315 Unit VI:
• Implemented unsupervised customer transaction segmentation using K-Means with StandardScaler (Silhouette Score: 0.2822).
• Demonstrated the empirical reality of financial spending estimation using Linear Regression ($R^2 \approx 0$, RMSE = ₹28,485.82).
• Benchmarked four comparative classifiers (Logistic Regression, Decision Tree, Linear SVM, Gaussian Naive Bayes), achieving 94.88% accuracy and 0.9238 F1-score, validated via a dynamic Confusion Matrix.

Future recommendations for enterprise deployment include integrating Apache Spark Structured Streaming with Apache Kafka for real-time transaction scoring and implementing minority class oversampling (SMOTE) to enhance fraud recall.

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